HW#6

Homework Report

Course	ECG 762
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Problem 01 6.1

If $x[n] = Ar^n + w[n]$ for $n = 0,1, \dots, N-1$, where A is an unknown parameter, r is a known constant, and $w[n]$ is zero mean white noise with variance σ^2, find the BLUE of A and the minimum variance. Does the minimum variance approach zero as $N \to \infty$?

Solution:

The observation model is

$$x[n] = Ar^n + w[n], n = 0,1, \dots, N-1.$$

Since the noise has zero mean,

$$E(\mathbf{x}) = A\mathbf{s},$$

where

$$\mathbf{s} = [1 \quad r \quad r^2 \quad \cdots \quad r^{N-1}]^T.$$

The noise is white with variance σ^2, so its covariance matrix is

$$\mathbf{C} = \sigma^2\mathbf{I}.$$

Using the scalar BLUE formula,

$$\begin{aligned}\hat{A}_{\text{BLUE}} &= \frac{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{x}}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}} \\ &= \frac{\mathbf{s}^T\mathbf{x}}{\mathbf{s}^T\mathbf{s}} \\ &= \frac{\sum_{n=0}^{N-1} r^n x[n]}{\sum_{n=0}^{N-1} r^{2n}}.\end{aligned}$$

This estimator is unbiased because

$$E\left(\hat{A}_{\text{BLUE}}\right) = \frac{\sum_{n=0}^{N-1} r^n Ar^n}{\sum_{n=0}^{N-1} r^{2n}} = A.$$

Its minimum variance among linear unbiased estimators is

$$\text{Var}\left(\hat{A}_{\text{BLUE}}\right) = \frac{1}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}} = \frac{\sigma^2}{\sum_{n=0}^{N-1} r^{2n}}.$$

For real r, the geometric-series formula gives

$$\text{Var}\left(\hat{A}_{\text{BLUE}}\right) = \begin{cases}\sigma^2 \dfrac{1-r^2}{1-r^{2N}}, & r^2 \neq 1, \\ \dfrac{\sigma^2}{N}, & r^2 = 1.\end{cases}$$

The limiting behavior depends on $|r|$:

- If $|r| < 1$, then $r^{2N} \to 0$, and

$$\lim_{N\to\infty} \mathrm{Var}\left(\hat{A}_{\mathrm{BLUE}}\right) = \sigma^2(1-r^2) > 0.$$

- If $|r| = 1$, then

$$\mathrm{Var}\left(\hat{A}_{\mathrm{BLUE}}\right) = \frac{\sigma^2}{N} \to 0.$$

- If $|r| > 1$, then

$$\mathrm{Var}\left(\hat{A}_{\mathrm{BLUE}}\right) = \sigma^2 \frac{r^2-1}{r^{2N}-1} \to 0.$$

Thus, the minimum variance approaches zero if and only if $|r| \geq 1$, assuming $\sigma^2 > 0$. When $|r| < 1$, the signal decays, and the accumulated signal energy remains finite, leaving a positive limiting variance.

Problem 02 6.4
The observed samples $\{x[0], x[1], \ldots, x[N-1]\}$ are IID according to the following PDFs:
a. Laplacian

$$p(x[n];\mu) = \frac{1}{2}\exp[-|x[n]-\mu|]$$

b. Gaussian

$$p(x[n];\mu) = \frac{1}{\sqrt{2\pi}}\exp\left[-\frac{1}{2}(x[n]-\mu)^2\right].$$

Find the BLUE of the mean μ in both cases. What can you say about the MVU estimator for μ ?

Solution:
In both cases, the observations have mean μ and are IID. Therefore,

$$E(\mathbf{x}) = \mu\mathbf{1}, \mathbf{C} = v\mathbf{I},$$

where v is the variance of an individual observation.
Applying the BLUE formula gives

$$\hat{\mu}_{\mathrm{BLUE}} = \frac{\mathbf{1}^T\mathbf{C}^{-1}\mathbf{x}}{\mathbf{1}^T\mathbf{C}^{-1}\mathbf{1}} = \frac{1}{N}\sum_{n=0}^{N-1} x[n] = \bar{x}$$

Its variance is

$$\mathrm{Var}(\hat{\mu}_{\mathrm{BLUE}}) = \frac{v}{N}.$$

Hence, the sample mean is the BLUE in both cases, although its variance differs.
(a) Laplacian distribution

For

$$p(x[n]; \mu) = \frac{1}{2} e^{-|x[n]-\mu|},$$

the distribution is symmetric about μ, and

$$E(x[n]) = \mu.$$

Its variance is

$$\begin{aligned} \mathrm{Var}(x[n]) &= \frac{1}{2}\int_{-\infty}^{\infty} z^2 e^{-|z|} dz \\ &= \int_{0}^{\infty} z^2 e^{-z} dz \\ &= 2 \end{aligned}$$

Therefore,

$$\hat{\mu}_{\mathrm{BLUE}} = \frac{1}{N}\sum_{n=0}^{N-1} x[n], \mathrm{Var}(\hat{\mu}_{\mathrm{BLUE}}) = \frac{2}{N}.$$

However, being BLUE does not establish that the sample mean is MVU. A nonlinear unbiased estimator may have a smaller variance.

For example, when $N = 3$, the sample median

$$M = \mathrm{median}\{x[0], x[1], x[2]\}$$

is unbiased by symmetry and has variance

$$\mathrm{Var}(M) = \frac{23}{36} < \frac{2}{3} = \mathrm{Var}(\bar{x}).$$

Thus, for $N = 3$, the sample mean is BLUE but is not MVU. This comparison does not, by itself, establish that the median is MVU.
(b) Gaussian distribution

Here,

$$x[n] \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1),$$

SO

$$\hat{\mu}_{\text{BLUE}} = \frac{1}{N}\sum_{n=0}^{N-1} x[n], \text{Var}(\hat{\mu}_{\text{BLUE}}) = \frac{1}{N}.$$

The Fisher information for μ in the N observations is

$$I(\mu) = N.$$

Consequently, the Cramér-Rao lower bound for any unbiased estimator is

$$\text{Var}(\hat{\mu}) \geq \frac{1}{N}.$$

The sample mean is unbiased and attains this bound. Therefore, in the Gaussian case, it is both BLUE and MVU:

$$\hat{\mu}_{\text{MVU}} = \hat{\mu}_{\text{BLUE}} = \frac{1}{N}\sum_{n=0}^{N-1} x[n].$$

Problem 03 6.6
In this problem we extend the scalar BLUE results. Assume that $E(x[n]) = \theta s[n] + \beta$, where θ is the unknown parameter to be estimated and β is a known constant. The data vector **x** has covariance matrix **C**. We define a modified linear (actually affine) estimator for this problem as

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + b.$$

Prove that the BLUE is given by

$$\hat{\theta} = \frac{\mathrm{s}^T\mathbf{C}^{-1}(\mathbf{x} - \beta\mathbf{1})}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}.$$

Also, find the minimum variance.

Solution:
Write the mean vector and the affine estimator as

$$E(\mathbf{x}) = \theta\mathbf{s} + \beta\mathbf{1}, \hat{\theta} = \mathbf{a}^T\mathbf{x} + b.$$

Assume that **C** is positive definite and $\mathbf{s} \neq \mathbf{0}$.
The expectation of the estimator is

$$\begin{aligned} E(\hat{\theta}) &= \mathbf{a}^T E(\mathbf{x}) + b \\ &= \theta\mathbf{a}^T\mathbf{s} + \beta\mathbf{a}^T\mathbf{1} + b. \end{aligned}$$

For the estimator to be unbiased for every θ, we require

$$\mathbf{a}^T\mathbf{s} = 1, b = -\beta\mathbf{a}^T\mathbf{1}.$$

Thus, every unbiased estimator of the specified form can be written as

$$\hat{\theta} = \mathbf{a}^T(\mathbf{x} - \beta\mathbf{1}), \mathbf{a}^T\mathbf{s} = 1.$$

Since subtracting a known constant does not change the covariance,

$$\mathrm{Var}(\hat{\theta}) = \mathbf{a}^T\mathbf{C}\mathbf{a}.$$

We therefore minimize

$$\mathbf{a}^T\mathbf{C}\mathbf{a}$$

subject to

$$\mathbf{a}^T\mathbf{s} = 1.$$

Introduce the Lagrangian

$$L(\mathbf{a}, \lambda) = \mathbf{a}^T\mathbf{C}\mathbf{a} - 2\lambda(\mathbf{a}^T\mathbf{s} - 1).$$

Differentiating with respect to a and setting the gradient to zero yields

$$2\mathbf{C}\mathbf{a} - 2\lambda\mathbf{s} = \mathbf{0},$$

so

$$\mathbf{a} = \lambda\mathbf{C}^{-1}\mathbf{s}.$$

Applying the unbiasedness constraint gives

$$1 = \mathbf{a}^T\mathbf{s} = \lambda\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}.$$

Hence,

$$\lambda = \frac{1}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}},$$

and the optimal coefficient vector is

$$\mathbf{a}_{\text{opt}} = \frac{\mathbf{C}^{-1}\mathbf{s}}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}.$$

Because $\mathbf{C}$ is positive definite, the objective is strictly convex, so this solution is the global minimum. Substituting the optimal coefficients into the estimator gives

$$\hat{\theta}_{\text{BLUE}} = \frac{\mathbf{s}^T\mathbf{C}^{-1}(\mathbf{x} - \beta\mathbf{1})}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}.$$

The corresponding constant term is

$$b_{\text{opt}} = -\beta \frac{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{1}}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}}.$$

Finally, the minimum variance is

$$\begin{aligned} \text{Var}\left(\hat{\theta}_{\text{BLUE}}\right) &= \mathbf{a}_{\text{opt}}^T \mathbf{C} \mathbf{a}_{\text{opt}} \\ &= \frac{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{C} \mathbf{C}^{-1} \mathbf{s}}{(\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s})^2} \\ &= \frac{1}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}}. \end{aligned}$$

The known offset β is removed before estimation and therefore does not affect the minimum variance.